

TAWAU, Malaysia

WMO: 964810

Lat: 4.313N

Lon: 118.122E

Elev: 57

StdP: 14.67

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	72.2	73.0	68.0	103.1	79.6	69.5	108.5	77.9	12.5	85.1	11.7	85.4	2.7	330	0.301

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	12.9	91.3	79.0	90.0	79.2	89.6	79.3	82.0	87.8	81.3	87.5	80.8	87.2	8.5	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
80.5	158.6	87.0	79.2	151.9	85.8	79.0	150.7	85.6	46.1	87.4	45.1	88.8	44.6	87.6	89.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
12.2	11.2	9.9	DB	69.2	93.7	2.8	2.7	67.2	95.7	65.6	97.3	64.0	98.8	62.0	100.8
			WB	67.2	84.8	2.8	2.5	65.2	86.6	63.5	88.1	61.9	89.5	59.8	91.3

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	81.2	80.6	80.9	81.5	81.7	81.9	81.3	80.8	80.9	81.2	81.4	81.4	81.0
	DBStd	1.44	1.52	1.58	1.60	1.27	1.34	1.44	1.30	1.34	1.36	1.34	1.16	1.46
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	11394	950	866	975	951	988	938	955	959	935	973	942	962
	CDD65	5919	485	446	510	501	523	488	490	494	485	508	492	497
	CDH74	55856	4181	3990	4842	4973	5314	4721	4597	4723	4675	4802	4549	4489
	CDH80	19064	1309	1335	1722	1756	1888	1616	1519	1600	1638	1682	1541	1458
(14) Wind	WSAvg	3.8	3.6	4.0	4.1	3.8	3.6	3.7	3.8	4.1	3.9	3.7	3.4	3.4
(15) Precipitation	PrecAvg	88.0	8.9	6.3	7.1	6.3	7.8	7.2	6.9	6.7	6.1	8.1	8.0	8.7
	PrecMax	118.3	13.3	13.7	14.7	12.7	11.5	14.2	12.8	10.0	10.7	14.5	11.1	13.6
	PrecMin	59.8	4.3	1.1	1.8	1.4	5.7	1.9	3.8	2.5	1.4	3.6	4.2	3.5
	PrecStd	15.4	2.6	3.3	3.1	2.4	1.6	2.1	1.8	1.8	2.2	2.5	1.9	2.5
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	90.3	91.6	92.9	91.4	91.3	91.0	89.9	90.0	90.7	91.5	91.1	91.0
		MCWB	78.7	78.1	77.0	79.3	80.6	80.0	79.4	79.1	79.3	79.3	79.8	79.8
	2%	DB	89.1	89.8	90.8	89.8	89.8	89.6	88.6	89.2	89.5	89.8	89.6	89.5
		MCWB	78.5	78.3	77.7	79.3	80.1	79.7	79.2	79.1	79.3	79.5	79.8	79.6
	5%	DB	87.8	88.2	89.3	88.8	89.1	88.2	87.8	88.0	88.2	88.7	88.2	88.0
		MCWB	78.4	78.3	78.2	79.2	79.9	79.5	79.0	79.1	79.1	79.5	79.6	79.4
	10%	DB	86.3	87.1	87.7	87.7	87.8	87.4	86.6	86.9	87.5	87.7	87.5	87.0
		MCWB	78.1	78.1	78.3	79.2	79.7	79.3	78.6	78.7	78.9	79.4	79.4	79.1
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	81.5	82.9	81.4	81.9	82.8	82.1	81.7	81.3	81.6	82.3	82.5	82.0
		MCDB	87.5	87.1	88.0	87.8	88.2	88.1	86.9	87.2	87.7	88.5	88.1	88.3
	2%	WB	80.2	80.1	80.1	80.9	81.6	80.9	80.4	80.1	80.5	80.9	81.0	80.9
		MCDB	86.4	86.7	87.0	87.3	87.5	87.2	86.6	86.8	87.4	87.6	87.3	87.1
	5%	WB	79.3	79.2	79.4	80.1	80.8	80.2	79.6	79.5	79.8	80.1	80.2	80.1
		MCDB	85.4	85.7	86.2	86.6	87.0	86.4	85.8	86.1	86.5	86.7	86.5	86.2
	10%	WB	78.7	78.5	78.8	79.5	80.0	79.5	78.9	79.0	79.1	79.4	79.5	79.3
		MCDB	84.7	84.9	85.5	85.8	86.1	85.6	85.2	85.4	85.6	85.8	85.6	85.1
(35) Mean Daily Temperature Range	5% DB	MDBR	12.7	13.3	13.7	13.1	12.9	12.8	12.7	12.9	13.3	13.5	13.4	12.8
		MCDBR	14.4	15.2	15.8	14.4	14.2	14.1	13.9	14.1	14.5	15.0	14.6	14.3
		MCWBR	6.7	6.8	6.7	6.5	6.7	6.7	6.7	6.7	6.8	7.2	7.0	6.8
	5% WB	MCDBR	13.2	13.8	14.1	13.4	13.3	13.3	13.2	13.4	14.0	14.1	13.9	13.4
		MCWBR	7.0	7.0	6.9	6.8	7.0	6.9	6.9	6.9	7.1	7.4	7.3	7.0
(40) Clear-Sky Solar Irradiance	taub	0.399	0.405	0.406	0.411	0.402	0.404	0.409	0.451	0.466	0.484	0.409	0.396	(40)
	taud	2.529	2.491	2.484	2.464	2.512	2.514	2.497	2.354	2.303	2.249	2.485	2.538	(41)
	Ebn at Noon	293	293	291	284	280	276	276	269	270	266	288	292	(42)
	Edh at Noon	34	36	36	36	34	33	34	40	43	45	35	33	(43)
(44) All-Sky Solar Radiation	RadAvg	1468	1641	1704	1751	1665	1548	1583	1653	1651	1588	1544	1493	(44)
	RadStd	125	166	161	135	97	70	97	78	101	97	102	111	(45)

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	+0.31	N/A	N/A	+0.65	+0.85	+0.76	N/A	N/A	+218	+218
(47) Regional (2 neighbors)	+0.29	N/A	N/A	+0.59	+0.72	+0.54	N/A	N/A	+169	+169

Nomenclature: See separate page